CHUHAN ZHANG

(+86) 137-9937-5365 · czhan251@jh.edu · chuhandanielzhang.github.io

RESEARCH INTERESTS

I am interested in dynamic robots across locomotion and manipulation, at the intersection of *model-based control* and *reinforcement learning*. As a full-stack roboticist, I build complete systems, from mechanism and embedded real-time control to sim-to-real learning, to make robots robust and agile on real hardware.

EDUCATION

- **Johns Hopkins University** — Aug. 2026 – Dec. 2027
  M.S.E. in Robotics (Medical Robotics track)
- **Technion – Israel Institute of Technology (GTIIT)** — Sep. 2022 – Oct. 2026
  B.S. in Mechanical Engineering and Robotics (GPA: 88.0/100, ≈3.7/4.0)
- **The Chinese University of Hong Kong (CUHK)** — Jul. 2025 – Sep. 2025
  Visiting Research Student

PUBLICATIONS

1. **Chuhan Zhang**, Hongbo Zhang, Yanlin Chen, Yunxi Tang, Yun-Hui Liu, Mingyi Liu, Xiangyu Chu. Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation. IEEE International Conference on Automation Science and Engineering (CASE), 2026. (Accepted) [arXiv] [Video]

2. Tongchen Lin†, Yanqiu Zheng†, **Chuhan Zhang**, Ruigang Chen, Yizhar Or, Mingyi Liu. Dynamics, Stability, and Energy Efficiency of an Energy-Recycling Rimless Wheel with Spring-Clutch Legs. Robotica, 2026. (Under review) [arXiv] [Video]

RESEARCH EXPERIENCE

Tactile Simulation for Contact-Rich Manipulation — Summer 2026 – Present
Research Assistant, Johns Hopkins University — Advisor: Prof. Krishna Murthy Jatavallabhula

- Developing tactile-sensor simulation to study how tactile feedback affects contact-rich manipulation.
- Connecting simulated contact signals to visuotactile policy learning for manipulation tasks.

Propeller-Assisted 3D Hopping Robot — Jul 2025 – Present
Research Assistant, CUHK — Advisors: Prof. Xiangyu Chu **Model-based control (first author, CASE 2026):** — Jul 2025 – Mar 2026

- Built Pro-OMEGA, a propeller-assisted 3D monopedal hopping robot pairing an active 3-RSR parallel leg with a trunk-mounted tri-rotor.
- Proposed a single-rigid-body **Hierarchical Force Allocation (HFA)** controller: the leg produces the stance wrench via a friction- and torque-constrained **QP**, and the tri-rotor supplies the residual attitude moment in stance and full roll/pitch authority in flight.
- Built the **real-time control stack** (500 Hz LCM loop, 3-RSR inverse kinematics, Raibert-style foot placement) on a Raspberry Pi and flight controller.
- Validated on hardware with one fixed controller: continuous 3D hopping at ∼1.6 m/s with **abrupt start-stop** and velocity reversals, a grass-to-wood terrain transition, and recovery from impulsive pushes within 1–3 hops.

Reinforcement learning: — Jul 2025 – Present

- Trained **PPO** hopping policies for the 3-RSR parallel leg in **Isaac Gym** and **MuJoCo MJX** across thousands of GPU-parallel environments.

- Compared an equivalent serial-leg model and a native closed-chain MJX model for **sim-to-real** of the closed kinematic chain.
- Designed an **encoder-decoder policy** for flight-phase underactuation, added **domain randomization** and **actuator-delay modeling**, and benchmarked against the model-based QP controller on real-robot logs.

Multimodal platform: Mar 2026 – Present

- Extended Pro-OMEGA into a multimodal platform that robustly switches among hopping, mobile, and manipulation modes outdoors, with onboard LiDAR SLAM for mapping and navigation.

Energy-Efficient Legged Robot via Energy Recycling Nov 2023 – Jul 2025
Research Assistant, GTIIT Advisor: Prof. Mingyi Liu

- Modeled **hybrid dynamics** of a passive walker using **Lagrangian methods** and analyzed stability with **Poincaré maps** via MATLAB.
- Designed and prototyped a **spring-clutch mechanism** to recycle collision energy, achieving lower energy loss.
- Performed **motion-capture experiments** to validate the model, achieving a **60% reduction in transport cost** to 0.02.

COMPETITION EXPERIENCE

RoboMaster University League (RMUL) Oct 2024 – Mar 2025
Mechanical Adviser for Kongfu Team, Technion, Shantou, China

- Co-founded and developed Technion, Shantou, China's first robotics club and designed and developed the RoboMaster Standard Robot.
- Constructed a robot with omnidirectional mobility, slope-traversing suspension, and an integrated gimbal system.
- Developed a visual automatic targeting system for the gimbal firing mechanism.
- Led the team to win Second Prize in the China RoboMaster University League.

VEX-EDR Competition & Innovation Competition Sep 2015 – Dec 2020
Team Leader, Fuzhou No. 8 Middle School, China

- Co-founded the middle school Team 9698B, leading the design, build, and programming of a holonomic ball-intake and flywheel-shooter robot for the VEX "Nothing But Net" season.
- Founded the high school Team 96666A and mentored two new members through CAD, fabrication, and autonomous control to develop a cube-stacking tray robot for the VEX "Tower Takeover" season.

HONORS & AWARDS

- 2026 Israel Innovation Prize (Undergraduate Category).
- Second Prize, China RoboMaster University League, 2025.
- GTIIT Dean's List Academic Scholarship, 2022–2026 (top 15%).
- Gold Award, VEX Asia-Pacific Robotics Championship (High School Group), 2019.
- China Youth Science and Technology Innovation Award, 2018.
- Champion, VEX Robotics World Championship (Middle School Group), 2016.

TECHNICAL SKILLS

Programming: C++, Python, MATLAB
Tools: Isaac Gym, MuJoCo MJX, ROS2, LCM, SolidWorks, AutoCAD, Blender, Adobe Illustrator
Test Scores: TOEFL iBT (2026): 5.5/6.0